

The Latent Relay is a Prefix, Not a Message: Interpreting Inter-Agent KV Communication

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Abstract

Latent multi-agent systems such as LatentMAS replace text messages with a shared key-value (KV) cache: silent Planner, Critic, and Refiner agents grow a relay of latent steps, and only a Judger decodes. The design is motivated by the claim that this relay is a *richer native communication channel* than language. We treat that claim as an interpretability question and intervene on the cache after the upstream agents finish.

Two complementary tests argue that the $K=40$ tape is mostly a content-insensitive prefix. *Swap*: a wrong-item cache matches a right-item cache for Judger accuracy under matched token budgets. *Delete*: after a full $K=40$ rollout, keeping attention sinks plus a small top- $\|K\|$ subset of slots leaves GSM8K and MATH-500 accuracy in the same band as the uncompressed system (96.3% vs. 95.7%; 85.7% vs. 83.7%, three seeds, paper sampling). Aggressive deletion (keep-16) does not collapse GSM8K accuracy but makes the Judger longer, so a little structure must remain. Eviction is not a competing multi-agent algorithm and is not a substitute for running fewer latent steps; it is a length ablation of a *finished* $K=40$ cache. The systems side-effect is smaller Judger prefixes and lower batched KV memory, not skipped upstream compute.

We recommend reading LatentMAS’s native channel as a decoding-mode scaffold, not as an instance-specific message.

1 Introduction

Multi-agent prompting lets models play specialized roles and exchange text (Du et al., 2023; Wu et al., 2023; Wei et al., 2022). Zou et al. (2025) push the messages *inside* the transformer: three upstream agents run a fixed number K of latent forward passes, emit no tokens, and append to a shared KV cache (past_key_values). A final Judger attends to that cache and generates the answer. They argue that this *latent collaboration* is

richer than language and, in their main tables, operate at a moderate latent budget on the order of $K=40$.

If the relay were an instance-specific reasoning transcript, two things should fail. Replacing it with another problem’s cache should hurt. Deleting most of its slots after a long rollout should hurt. We find the opposite on the settings we can measure carefully.

What this paper is. An analysis of the latent channel in a verified sequential LatentMAS fork (Qwen3-14B; Yang et al., 2025). Contribution type: model analysis, not a new multi-agent method.

What this paper is not. We do not claim a compressor that beats LatentMAS on accuracy or end-to-end latency. We do not claim that LatentMAS’s latent-step sweep (their reported peak near $K \in [40, 80]$) is false: we did not rerun that full grid. We do not claim that running $K=10$ is the same algorithm as compressing a $K=40$ cache (§6).

Interventions. (i) Cache *swap*: decode the Judger from another item’s relay. (ii) Cache *eviction*: after the same $K=40$ upstream pass, keep a few sink positions plus the largest key-norm slots and decode. Swap tests *content*. Eviction tests *length* of an already-built tape. Together they support a simple picture: the native channel is a long, noisy prefix that puts the Judger into a working decoding mode; it is not a tight code for the answer.

2 LatentMAS, in the form we study

Pipeline. Sequential LatentMAS runs three silent agents then a text Judger: $\text{Planner}_K \rightarrow \text{Critic}_K \rightarrow \text{Refiner}_K$ grow a shared relay C , and only the Judger decodes. Each silent agent realigns its last-layer state with W_a and runs K extra forwards, appending KV slots. Figure 1 marks the intervention: *after* C exists.

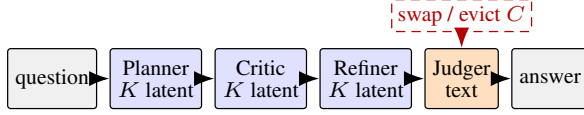


Figure 1: Sequential LatentMAS. Silent agents grow a shared KV relay C ; only the Judger decodes. We intervene on C immediately before Judger decode.

Fork. Code is a byte-for-byte copy of Gen-Verse/LatentMAS at commit 9a9e4d3 (24/24 file SHAs). At $K=40$, sequential HF decode, Qwen3-14B, a one-seed $n=150$ check sits within 1–2.5 points of their published GSM8K / ARC-Challenge / MedQA table means (Appendix A). Eviction is layered on that path: we do not reimplement the agents.

Default operating point. Unless noted, $K=40$, Judger temperature 0.6 and top- p 0.95 (their sampling), one decode per item. GSM8K (Cobbe et al., 2021), MATH-500 (Hendrycks et al., 2021; Lightman et al., 2023), GPQA-Diamond (Rein et al., 2023), AIME 2024/2025. Seeds {42, 123, 7}; $n=100$ except AIME ($n=30$, full contest set; AIME 2025 missing seed 7).

3 Two knives on the same object

Let C_x be the relay produced by Planner/Critic/Refiner on question x . The Judger implements $p_\theta(y | x, C)$.

Swap (content). Decode from $C_{x'}$ for a different same-task x' , or from no cache. If C carried x -specific intermediate reasoning, accuracy should track whether $x' = x$. Diagnostics use greedy Judger decode and matched generation budgets $T_{\max}$ (MedQA, Qwen3-14B, $n=40$).

Eviction (length). After building C_x at $K=40$, keep $s=4$ attention-sink slots per layer (Xiao et al., 2023) plus the remaining budget of positions with largest ℓ_2 key norm (Zhang et al., 2023). GSM8K keep-64; MATH/GPQA/AIME keep-128. The upstream pass is *identical* to full LatentMAS; only the Judger prefix changes. This is not early-stopping. Early-stopping at $K=10$ produces the sequence $(a_{1:10}, b_{1:10}, c_{1:10})$. Eviction *selects* from $(a_{1:40}, b_{1:40}, c_{1:40})$. Section 6 records both only so a reader cannot confuse them.

What eviction is allowed to show. If the extra thirty latent steps stored a dense, necessary message, dropping most slots of a finished $K=40$

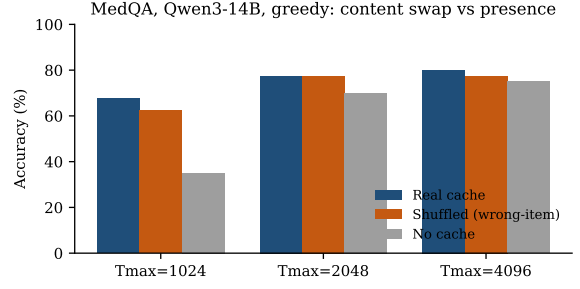


Figure 2: Swap vs. presence on MedQA (greedy, $n=40$). A wrong-item cache tracks the real cache; no cache hurts mainly when the Judger is short of tokens.

cache should collapse accuracy. If it does not, those steps are a noisy / redundant tape *at this operating point*. That is an interpretability result. Smaller KV and slightly faster Judger attention are side-effects (§7).

4 The channel ignores instance content

Figure 2 restates the swap diagnostic. At $T_{\max}=1024$, removing the cache drops MedQA from 67.5% to 35.0%: *presence* of a latent prefix matters under a tight budget. Replacing it with another item’s cache stays close to the real cache (62.5% vs. 67.5%). At 2048 and 4096 tokens the Judger can re-solve; real and shuffled are tied, and the gap to no-cache shrinks. GSM8K at $T_{\max}=1024$ is the same shape (real 93.3%, shuffled $\approx 93\%$, none 80%).

Linear and MLP probes of the gold answer from per-agent hidden states are near chance (four-way MedQA, ≈ 0.28 – 0.33 vs. 0.25). A correctness probe is weak upstream and stronger at the Judger. Three attempts to *steer* the latent channel for accuracy (diff-of-means residuals, KV-cache steering of handoff columns, a learned rank objective) do not beat an unsteered baseline; Judger-only concision steering *does* change length (Chen et al., 2025; Li et al., 2025). We treat those as converging negatives, not as this paper’s headline, and omit them from the main tables.

5 Most of a finished $K=40$ tape is droppable

Table 1 and Figure 3 report eviction under *their* sampling. Full $K=40$ vs. the same cache after eviction, paired on the same items.

Easy math. GSM8K is saturated either way. MATH-500 moves by two points. If the uncom-

Task	$n \times \text{seeds}$	$K=40$	Evict	Judger s
GSM8K keep-64	100×3	96.3	95.7	14.4 / 13.0
MATH-500 keep-128	100×3	85.7	83.7	34.2 / 31.8
GPQA keep-128	100×3	53.7	48.7	87.6 / 65.8
AIME24 keep-128	30×3	36.7	32.2	151 / 144
AIME25 keep-128	30×2	35.0	31.7	155 / 144

Table 1: Mean accuracy (%) and mean Judger seconds (full / evict). Relay storage is ~ 100 MB vs. 10 MB (GSM8K keep-64) or 20 MB (keep-128). Upstream Planner/Critic/Refiner times match by construction (≈ 1 s each).

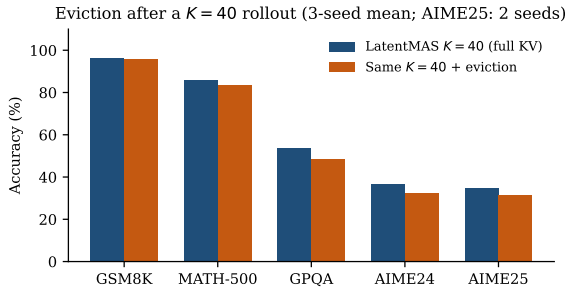


Figure 3: Accuracy after deleting most KV slots of a $K=40$ relay. Easy math holds; GPQA/AIME dip. This is a length ablation, not a new MAS.

pressed tape were a necessary workspace for the answer, keep-64/128 should not look like this.

Harder tasks. GPQA drops about five points; AIME about four, on $n=30$ where one item is 3.3 points. We do not spin a five-point GPQA dip into “free compression.” We also do not need a collapse to conclude redundancy: most of the tape can go and the system still answers in the same ballpark.

The keep-16 knee. On GSM8K with the same 0.6/0.95 protocol ($n=50$, seed 42), keep-16 ties full LatentMAS at 98% accuracy but the Judger runs *longer* (+3 s, $+\approx 120$ tokens). Deleting almost everything preserves the answer and wrecks the short-CoT mode. The scaffold is coarse, not empty.

6 Eviction is not $K=10$

A reviewer can ask: why not set latent_steps=10? That is a different system. It never computes steps 11–40. Eviction pays for those steps, then throws most of them away. Table 2 is therefore *not* a claim that eviction wins. On GPQA, stopping at 10 is more accurate than our key-norm subset of a $K=40$ cache. The row is here so the length-ablation is not misread as a hidden K -sweep.

Task	$K=40$ full	Evict	$K=10$ full
GSM8K	96.3	95.7	95.3
MATH-500	85.7	83.7	83.7
GPQA	53.7	48.7	59.0
AIME24	36.7	32.2	37.8

Table 2: Early-stop ($K=10$) vs. select-from- $K=40$. Different objects. $K=10$ is cheaper upstream (≈ 0.27 s vs. ≈ 1.0 s per silent agent) and is *not* our method.

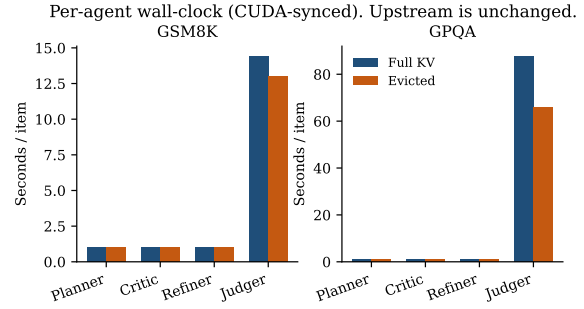


Figure 4: CUDA-synced per-agent time. The silent agents are a fixed tax at $K=40$; eviction only changes the Judger prefix.

We do *not* use Table 2 to contradict LatentMAS’s reported peak at $m \approx 40\text{--}80$. That figure is a multi-point sweep on their tasks and seeds. Two dots on GSM8K/MATH/GPQA, and AIME where one problem is three points, are not that experiment.

7 Systems side-effects, not the claim

Figure 4 splits wall-clock. Eviction cannot reduce Planner, Critic, or Refiner time: those forwards already happened. Judger time moves when traces are long (GPQA: 88 s $\rightarrow$ 66 s) and barely moves on GSM8K. Many AIME decodes hit an 8192-token cap, so Judger time is censored.

Figure 5 is the batch-memory plot. Qwen3-14B weights are ≈ 28 GB. One relay is ≈ 100 MB vs. 10 MB after keep-64. At Judger batch $B=64$ (short 16-token decode, GSM8K donor cache, H200), peak memory is 47 GB vs. 32 GB. Neither setting OOM. Compression is a batched-KV saving, not an OOM rescue on this GPU and not an accuracy result.

8 Related work

Latent and multi-agent reasoning. LatentMAS (Zou et al., 2025) is the system we analyze. Coconut (Hao et al., 2024) trains continuous latent thoughts inside one model; we study

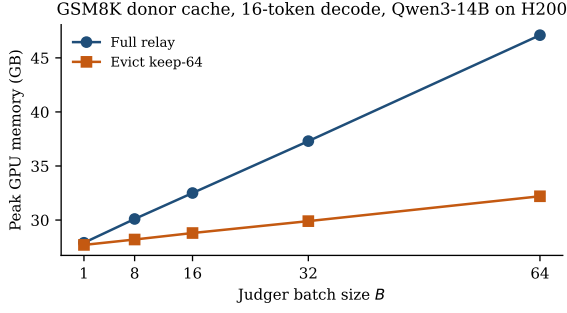


Figure 5: Peak memory vs. Judger batch size. Weights dominate; the relay is what scales with B .

a frozen multi-agent KV handoff, not a training recipe. Text MAS (Du et al., 2023; Wu et al., 2023) exchanges language; our question is whether replacing language with KV creates a message.

KV eviction. StreamingLLM attention sinks (Xiao et al., 2023) and H2O heavy hitters (Zhang et al., 2023) motivate the heuristic. We use them as a *delete test* on an inter-agent relay, not as an inference-engine contribution.

Analysis of internals. Probing, activation patching, and causal interventions (Belinkov and Glass, 2019; Meng et al., 2022; Geva et al., 2023) are the methodological family. Activation and KV steering (Turner et al., 2023; Rimsky et al., 2024; Chen et al., 2025; Li et al., 2025) supply the negative controls: the channel is hard to steer for accuracy; the text-emitting Judger is not.

9 Conclusion

The $K=40$ LatentMAS relay behaves like a long prefix that induces a Judger mode, not like a compact, instance-specific message. Swapping content does little; deleting most positions of a finished tape does little on easy math and only modestly more on GPQA. Naive eviction is the delete test that makes that length claim checkable, plus a smaller KV side-effect. The way to *improve* latent MAS, if one wants a methods paper, is to skip compute or replace the tape—not to celebrate building forty steps and throwing them away.

Limitations

One backbone (Qwen3-14B) and sequential LatentMAS only. Swap diagnostics are greedy, smaller n , MedQA/GSM8K. Eviction uses a single importance score; a better picker might keep more GPQA accuracy, which would *strengthen*

the “selection $\neq$ early-stop” distinction rather than the redundancy claim. AIME $n=30$ and frequent 8192 caps limit latency conclusions. LiveCodeBench was not run. We did not rerun LatentMAS Figure 8. Eviction never reduces upstream FLOPs. Results are analysis of one published system, not a universal theory of latent communication.

Ethics statement

This work analyzes a public multi-agent inference method on standard academic benchmarks. It does not release new models or user data. The main risk is over-interpreting a negative communication story as a reason to deploy shorter, under-evaluated agent stacks.

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Task ($K=40, n=150, 1$ seed)	Fork	Paper table
GSM8K	92.7	95.2
ARC-Challenge	94.7	95.6
MedQA	79.3	80.7

Table 3: Functional check of the 9a9e4d3 fork vs. published LatentMAS means (their 3-seed full sets). Gaps are consistent with subset and seed, not a different algorithm.

A Fork check

B Per-seed eviction accuracies

GSM8K $K=40$ /evict: 96/96, 97/96, 96/95.
 MATH-500: 86/86, 86/84, 85/81. GPQA: 52/48,
 52/44, 57/54. AIME24: 37/37, 37/27, 37/33
 (percent, $n=30$).